

TeamUp

Skill-Based Team Formation Platform for Projects

1. Introduction

1.1 Purpose and Scope

TeamUp – A Skill-Based Project Collaboration Platform is a software system designed to help students, developers, freelancers, researchers, and other professionals find suitable teammates based on their skills and collaborate on projects efficiently. Many individuals struggle to build project teams because they cannot easily identify people with the required expertise. TeamUp addresses this problem by providing a centralized platform where users can create profiles, list their skills, post projects, search for collaborators, and manage team activities.

The scope of the project includes user registration and authentication, profile management, skill management, project creation and management, collaboration requests, team formation, messaging, notifications, and an administrative panel for managing users and projects. The platform is intended for academic, personal, and professional collaboration, making it easier to connect skilled individuals and improve project success.

Target Audience:

- University students
- Software developers
- Freelancers
- Designers
- Researchers
- Startup teams
- System administrators

1.2 Definitions, Acronyms, and Abbreviations

SRS – Software Requirements Specification | UI – User Interface | DB – Database | API – Application Programming Interface | CRUD – Create, Read, Update, Delete | Admin – System Administrator | User – Registered platform user | Java – Programming language used for development | MongoDB – NoSQL database used to store data | IDE – Integrated Development Environment

2. General Description

2.1 Project Perspective

TeamUp is a Skill-Based Team Formation and Project Collaboration Platform designed to help students, freelancers, and professionals connect with suitable teammates based on their skills and interests. The platform provides a centralized environment where users can create profiles, post projects, recruit team members, manage tasks, communicate, and evaluate team performance after project completion.

Unlike general social media or job portals, TeamUp focuses specifically on project collaboration by combining skill-based matching, team recruitment, project management, and member ratings within a single system. The platform aims to improve collaboration, reduce the time required to build effective teams, and increase the success rate of academic, freelance, and innovation-based projects.

2.2 Project Functions

User Profile:

- Create Profile
- Edit Profile
- Upload Profile Picture
- Add Skills
- Add Bio

- Add Experience
- Portfolio Links

Team Recruitment:

- Accept/Reject Applicants
- Invite Members
- View Team Members

Skill-Based Matching:

- Match users according to skills
- Show Recommended Members
- Filter by Skills
- Filter by Experience

Rating & Review:

- Rate Members
- Give Feedback
- View Ratings

Search & Filter:

- Search Users
- Search Projects
- Filter by Skills
- Filter by Category
- Filter by Experience

2.3 User Classes and Characteristics

- Admin – Manage users, projects, and system activities
- Team Leader – Create projects and manage team members
- Team Member – Join projects and collaborate with the team
- Visitor – View public information and register an account

2.4 Operating Environment

Hardware: Laptop, Internet Connection | Software: Java (JDK), MongoDB | Operating System: Windows

2.5 Design and Implementation Constraints

- Developed using Java
- MongoDB is used as the database
- Internet connection is required
- Users must log in to access platform features

2.6 Assumptions and Dependencies

Assumptions:

- Users provide valid information
- Stable internet connection is available

Dependencies:

- Java Runtime Environment (JRE/JDK)
- MongoDB Database
- Internet connectivity

3. Specific Requirements

3.1 Functional Requirements

- Users can register and log in securely
- Users can create and update their profiles
- Users can add and manage their skills
- Users can create, edit, and delete projects
- Users can search for projects and collaborators by skills
- Users can send, accept, or reject collaboration requests
- Team members can communicate through messaging
- Users can rate and review team members after project completion
- Admin can manage users and projects

3.2 Non-Functional Requirements

- Performance: Search results should load within 2 seconds
- Security: Passwords must be encrypted, and only authorized users can access protected features
- Usability: The interface should be simple, responsive, and user-friendly
- Reliability: The system should be available and store data accurately
- Scalability: The system should support adding new users and features in the future

3.3 External Interface Requirements

- User Interface: Login page, Dashboard, Profile page, Project page, and Messaging page
- Software Interface: Java application connected to a MongoDB database
- Communication Interface: Internet connection using HTTP/HTTPS

3.4 Data Requirements

- Store user accounts, profiles, skills, projects, collaboration requests, messages, and notifications in MongoDB
- Ensure secure data storage using password hashing and validation
- Allow efficient data retrieval, update, and backup

4. Study of Existing Work

The table below compares TeamUp with existing skill-based collaboration and team-formation systems identified in the literature and in the market.

Project Name	Used Technology	Functional Features	Limitations	Our Contribution
CampLink [1]	ReactJS, Node.js, Express, MongoDB	Service posting & discovery, skill-based matching (recommendation engine), real-time chat, rating & feedback, institutional profile verification	Built around freelance/service exchange within a single campus; no structured project-team formation, no task management module, no admin analytics panel	Provides structured project-based team formation instead of one-off service exchange — dedicated project creation, collaboration requests, in-project task management, and an admin panel
Collab Connect [2]	HTML, CSS, JavaScript, Node.js, Express.js, MySQL	Student profile creation, skill classification, search & filter, collaboration requests, basic communication module, project team formation	Limited to academic peer discovery; no post-project rating/review system, no task assignment/progress tracking, no file-sharing messaging; still at review/prototype stage	Extends peer discovery into full project lifecycle management — adds team rating & review after completion, in-app task tracking, one-to-one/group messaging with file sharing, and real-time notifications
Devpost [3]	Web-based, React, Node.js, MongoDB	Hackathon participation, project showcase, team formation, submission, portfolio	Focuses primarily on hackathon submissions; limited built-in project	Provides full project lifecycle management including project creation, collaboration requests, task

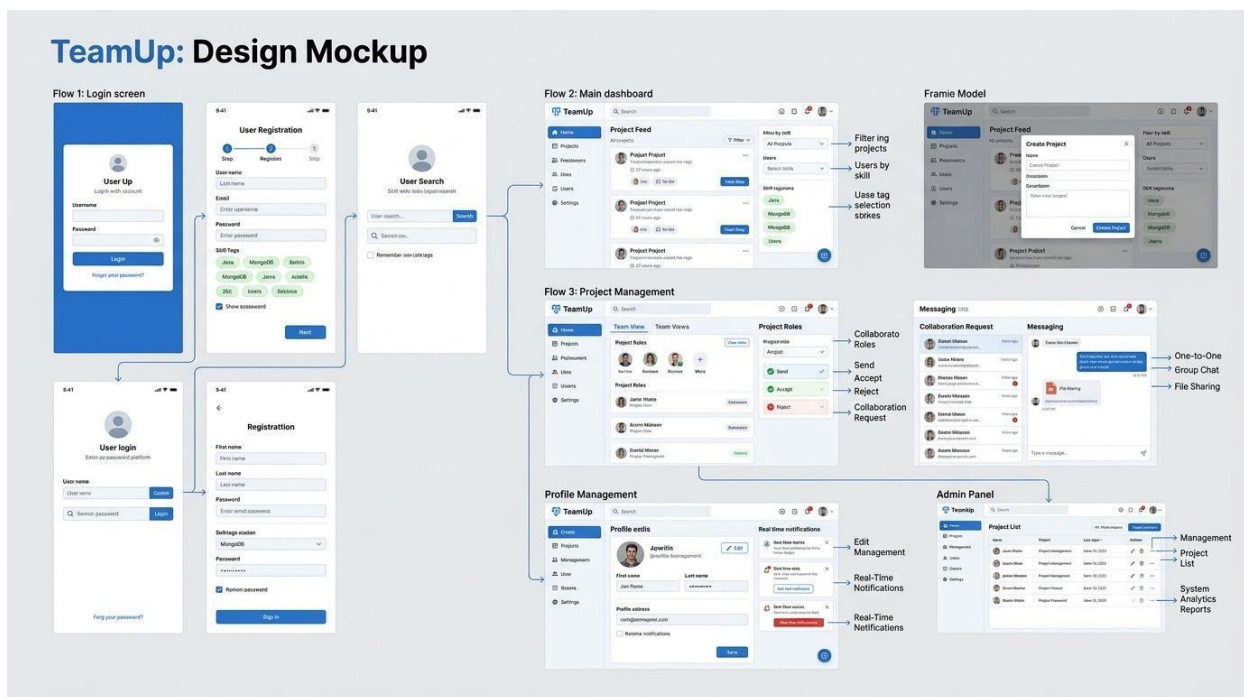
Project Name	Used Technology	Functional Features	Limitations	Our Contribution
			management and collaboration features	assignment, team communication, and ratings in one platform
Our Project — TeamUp	Core Java ,JDBC, MySQL/mongoDB	User registration & login, skill-based profiles, project creation, collaborator search, collaboration requests, task management, message board, notifications, team ratings, admin panel	Desktop application (future versions can include mobile/web support)	A complete collaboration platform built using Core Java that supports skill-based teammate matching, project management, communication, and team evaluation for students and professionals

Key Contributions of Our Project:

- Developed entirely using Core Java (Swing) and JDBC with MySQL
- Designed for both academic and professional project collaboration, not just hackathons
- Provides skill-based collaborator search
- Includes task management within each project
- Supports team messaging and notifications
- Implements a team rating and feedback system after project completion
- Offers an admin panel for managing users and projects

5. Proposed User Interface (UI) Design

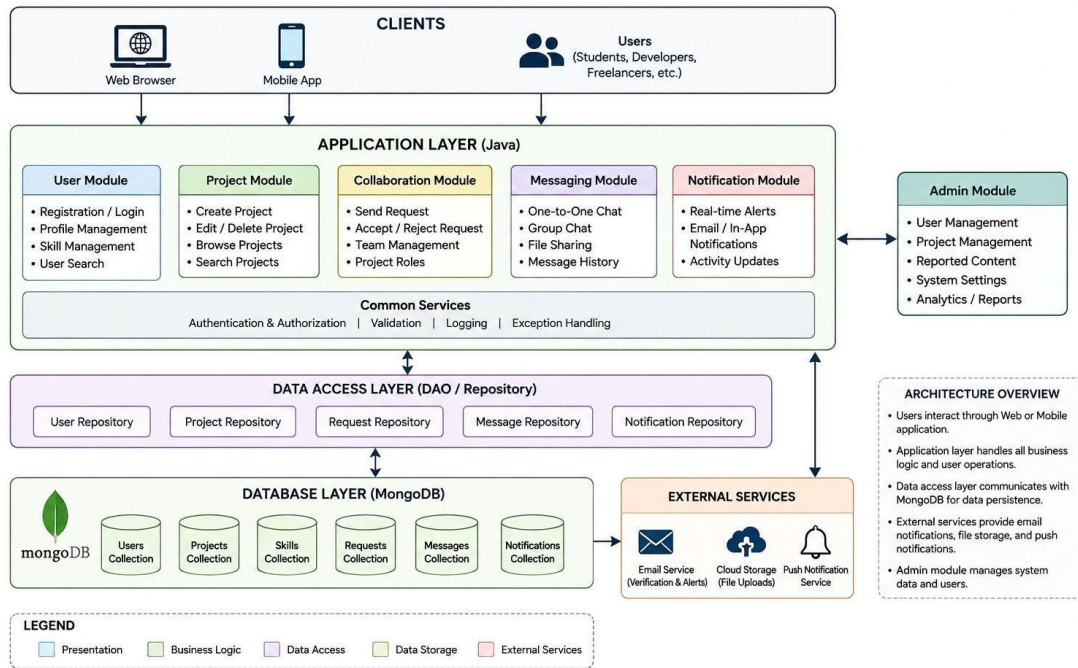
TeamUp: Design Mockup



6. System Architecture

4. SYSTEM ARCHITECTURE

High-level design of the Skill-Based Collaboration Platform



7. Other Supporting Information

7.1 Project Timeline

- Week 1 – Requirement Analysis
- Week 2 – System Design
- Week 3 – Database Design
- Week 4–5 – Development (User & Project Modules)
- Week 6 – Collaboration & Messaging Modules
- Week 7 – Testing and Bug Fixing
- Week 8 – Documentation and Final Presentation

7.2 Stakeholder List

- Project Supervisor
- Development Team
- System Administrator
- Students
- Developers/Freelancers
- End Users

7.3 User Documentation

Users can register, log in, create a profile, add skills, post projects, search for collaborators, send collaboration requests, communicate with team members, and manage their projects through a simple and user-friendly interface.

References

- [1] S. Khare, A. Shende, H. Gupta, N. Tiwari, K. Zade, and D. Kharabe, "CampLink: A Smart Platform for Collaborative Networking and Local Freelance Exchange Among College Students," International Journal of Engineering Research & Technology (IJERT), vol. 15, no. 1, Jan. 2026.
- [2] M. S. Taur, B. S. Deshmukh, and A. V. Khake, "A Skill Based Peer Discovery System: A Review," International Journal of Engineering Research & Technology (IJERT), vol. 15, no. 5, Jun. 2026.

[3] Devpost, Inc., "Devpost – Where Hackathons Happen," [Online]. Available: <https://devpost.com/>. [Accessed: Aug. 8, 2026].